

An Optimum Signal Detection Approach to the Joint ML Estimation of Timing Offset, Carrier Frequency and Phase Offset for Coherent Optical OFDM

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Abstract—Coherent orthogonal frequency-division multiplexing (OFDM) is one of the prime digital modulation techniques for present and future generations of wireless and optical communications. Accurate synchronization is the main obstacle to the implementation of a reliable OFDM receiver. We propose here a joint maximum likelihood (ML) estimator for the timing offset (TO), carrier frequency offset (CFO), and carrier phase offset (CPO) for coherent optical OFDM (CO-OFDM) motivated by the theory of ML signal detection. Our approach starts conceptually with the idea of first computing L replicas of the original OFDM spectrum, where L is a power of two and is assumed sufficiently large. This is done by padding $(L - 1)N$ zeros to the end of the N received noisy OFDM samples, where N is the number of OFDM subcarriers. By computing the LN -point discrete Fourier transform (DFT) of these LN time samples, we get L replicas of the DFT of the original OFDM spectrum, where each replica corresponds to the DFT for one hypothesized value of the CFO and the set of possible CFO values is $\{l/L\}_{l=0}^{L-1}$. We select the most probable replica by choosing the one that is at the minimum Euclidean distance from the original OFDM spectrum, which leads to a matched-filtering (MF) operation in the frequency domain in either a blind or a data-aided manner. Building on this MF concept, we then develop a joint ML estimator of the TO and CPO for each hypothesized value of the CFO. The novelty here is that the TO and the CPO are estimated efficiently as the frequency and phase of a complex sinusoid observed in noise, via either a time-domain or a

frequency-domain approach. The resulting joint CFO, CPO, and TO estimator is simpler than existing estimators, both conceptually and in implementation. A much simpler sequential approach in which we first decide on the CFO and then perform a joint TO and CPO estimation is also proposed. The performance loss of this sequential approach compared to the optimum joint approach is small at high signal-to-noise ratio (SNR). We obtain the performance of all our estimators via simulations, and show that they perform better when compared with the existing well-known estimators. Finally, we derive the Cramér–Rao lower bounds (CRLB) on the performance of our estimators, and show via simulations that our estimators for high SNR attain these performance lower bounds.

Index Terms—Carrier frequency offset, carrier phase offset, matched filter, maximum likelihood estimation, orthogonal frequency-division multiplexing (OFDM), timing offset.

I. INTRODUCTION

COHERENT optical orthogonal frequency-division multiplexing (CO-OFDM) has attracted much research attention due to its high spectral efficiency and its robustness to the chromatic dispersion (CD) and polarization mode dispersion (PMD) [1]. In a CO-OFDM system, a serial high-rate data stream is split into multiple parallel data streams with lower speed, each of which is modulated onto orthogonal subcarriers. The subcarriers are orthogonally overlapping with one another, leading to high spectral efficiency. However, synchronization errors in timing, frequency and phase can introduce both inter-symbol interference (ISI) and inter-carrier interference (ICI) that lead to severe performance degradation. The timing offset (TO) is caused by the propagation delay or the sample timing offset (frequency difference / phase offset) between the transmitter and receiver, and the carrier frequency offset (CFO) is induced by the frequency offset between the transmitter and receiver laser. The existence of a carrier phase offset (CPO) due to phase noise in the transmitter and receiver oscillators also causes signal constellation rotations in the subcarriers that can lead to bit error rate (BER) performance degradation. Therefore, achieving reliable synchronization is a major challenge for CO-OFDM systems.

Many approaches for CO-OFDM synchronization are available in the literature by now. The most promising approach to

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the design of synchronization algorithms for any digitally modulated signal is to treat the TO, CFO and CPO as unknown non-random parameters, and apply the theory of maximum likelihood (ML) estimation [2]–[7]. However, when applied to OFDM, this approach has turned out to be analytically intractable so far, and no solution is yet available that leads to a simple practical implementation. This is because these references, by and large, rely on solving the highly nonlinear likelihood equations that result from setting to zero the derivatives of the likelihood function with respect to each of these parameters. Various other synchronization approaches that may not be optimum with respect to any statistical criterion have also appeared. We present here a survey of the principal techniques available that are more commonly used and whose performance will form the benchmarks for comparison with the ML estimators we will derive. References [5], [8]–[18] have proposed methods to estimate the TO and CFO either jointly or independently. In [8], Schmidl and Cox proposed a training symbol-based method, by designing the first OFDM symbol with two identical halves and calculating the timing metric to estimate the TO. However, there is a plateau in the timing metric, whose length is equal to the length of the cyclic prefix (CP). When the signal-to-noise ratio (SNR) is low, the existence of the plateau can cause large timing estimation errors. To decrease the uncertainty of TO estimation, in [9], the authors modified Schmidl's method by dividing the training symbol into four identical parts, in which the first two parts have the same sign while the other two parts also have the same sign that are opposite to that of the first two parts. The resulting timing metric has a sharper roll-off, but still has a large timing-estimation error variance especially in ISI channels since there are two high side-lobes in the timing metric. In [10], Park utilized the conjugate symmetric sequence to design the training symbol structure, which leads to an even sharper peak, but the large side-lobes still exist which will affect the estimation accuracy significantly when the SNR is low. The OFDM signal is most sensitive to CFO, since the CFO causes the loss of orthogonality between the subcarriers, which introduces ICI to the system. The CFO normalized by the OFDM subcarrier spacing comprises an integral part and a fractional part. The work of [8] estimates the fractional part of the CFO by taking the correlation between the two identical halves in the training symbol and computing the phase of the correlation. However, the estimation of the integral part requires an exhaustive search, which leads to a high computational complexity. To ensure the CFO estimation stability under poor SNR conditions, in [12], the training symbol design is done in the frequency domain, and contains several data blocks. The CFO estimate is obtained via a two-step iterative operation in order to achieve better CFO estimation accuracy. In general, the data-aided (DA) approaches are easy for implementation; however, they require the insertion of a pilot symbol before each OFDM frame, which decreases the effective data transmission bit rate. Blind (BL) approaches can avoid this problem, and are common in applications. A deterministic blind approach in [13] makes use of a cost function that the authors approximated by a cosine function. The CFO estimate is obtained by solving three equations in the nonlinear cosine format that are obtained from three test values given by the

received signal samples. This approach is simple, but is sensitive to the SNR. Severe CFO estimation performance degradation may appear at low SNR.

Although one can propose various synchronization techniques [15]–[18], we feel that ultimately, the most reliable approach is to apply ML estimation theory. Our novel conceptual approach here begins by computing the OFDM spectrum at multiple hypothesized values of the CFO from the received noisy signal samples. The received time-domain signal of N samples is first zero-padded to the length of LN by padding $(L - 1)N$ zeros to the end of the signal. After performing the LN -point DFT, the spectrum obtained contains L noisy replicas of the original OFDM spectrum, each replica corresponding to one possible value of the CFO. The ML estimation of the CFO now amounts to a minimum-error-probability decision on which replica of the computed spectrum contains the original OFDM spectrum. From first principles in communication theory, the optimum decision is the one at the minimum Euclidean distance from the original OFDM spectrum, and the search for this optimum leads to a matched-filtering (MF) operation in the frequency domain. The MF concept obviates the need to solve a complicated likelihood equation, i.e., the equation obtained by equating to zero the derivative of the likelihood function with respect to the CFO. Solving the likelihood equation is the main obstacle so far in applying the ML principle to CFO estimation for CO-OFDM. The on-line implementation of the MF approach is efficient and conceptually simple. Both DA ML and BL ML estimation of CFO are achievable via the MF concept, and we will show that the estimation error variance of the DA ML estimator asymptotically reaches the Cramér-Rao lower bound (CRLB) as SNR increases, i.e., the estimator is asymptotically efficient.

Building on the MF concept, we develop a joint ML estimator for the CFO, CPO and the TO of an OFDM signal. For each discrete hypothesized value of the CFO, the receiver computes the joint ML estimates of the CPO and TO. The key novelty here is that we compute the latter joint estimates as the ML estimates of the frequency and phase of a complex sinusoid of constant amplitude observed in additive white Gaussian noise (AWGN). These ML estimates can be computed explicitly via the time-domain estimator in our work in [19], or the frequency-domain technique in [20] that involves a search for the peak of a periodogram. The MF CFO estimator coupled with the novel ML CPO and TO estimator leads to a much simpler receiver than existing ones. To reduce the implementation complexity, we propose a suboptimum sequential approach in which we first obtain the ML CFO estimate, followed by the joint ML estimation of CPO and TO corresponding to the CFO estimate obtained. The performance loss of this sequential approach compared to the optimum joint approach is small at high SNR. We obtain the performance of all our estimators via simulations, and show that they perform better when compared with the existing well-known estimators mentioned above. Finally, we derive the CRLBs on the performance of our estimators, and show via simulations that our estimators for high SNR attain these performance lower bounds.

The paper is organized as follows. Section II introduces the signal model. The proposed joint ML parameter estimator is

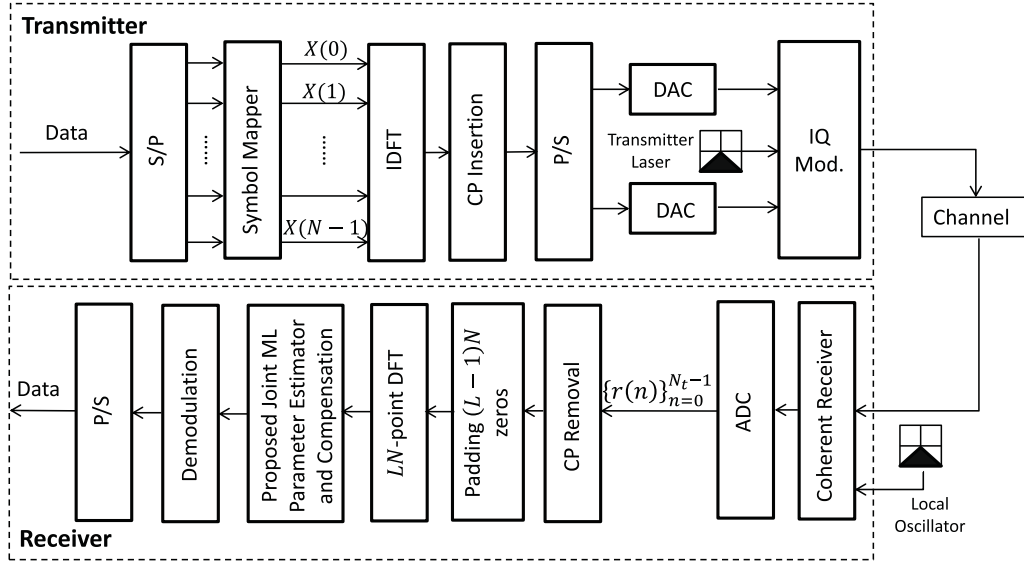


Fig. 1. Schematic diagram of a coherent OFDM system. S/P: serial to parallel, (I)DFT: (inverse) discrete Fourier transform, CP: cyclic prefix, DAC: digital to analog converter, IQ Mod.: IQ modulator. The value of N_t is equal to N plus the length of CP.

derived in Section III. Section IV shows a sequential parameter estimator with much simpler implementation. Section V derives the performance bounds. Section VI provides the simulation results and the discussion of the performance comparison. Finally, Section VII concludes the paper.

II. SIGNAL MODEL

The schematic diagram of a CO-OFDM system is shown in Fig. 1. According to Fig. 1, the n -th received time-domain OFDM sample $r(n)$ after removing the CP, with the appearance of TO, CFO and carrier phase noise (CPN), can be expressed as [1], [5]:

$$r(n) = x(n - \tau) \otimes h(n) e^{j[2\pi\epsilon n/N + \theta(n)]} + w(n), \quad (1)$$

where we have $n = 0, 1, \dots, N - 1$. Term N is the number of subcarriers, $x(n)$ is the n -th transmitted OFDM sample, which is obtained by taking the N -point inverse discrete Fourier transform (IDFT) of the data spectrum $\{X(k)\}_{k=0}^{N-1}$, where $X(k)$ is the data on the k -th subcarrier and can be expressed as [1]:

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}, k = 0, 1, \dots, N - 1. \quad (2)$$

Note that the TO denoted by τ is in the units of sample period (T_s) which is assumed to be within the range of $[0, N - 1]$. Term $h(n)$ is the channel impulse response, which in general represents the dispersive channel, including the CD and PMD. Term $w(n)$ is the n -th complex additive white Gaussian noise (AWGN) sample with mean 0 and variance σ^2 , and $\otimes$ denotes the convolution operation. Term ϵ is the normalized CFO which is equal to the value of actual CFO normalized by the OFDM subcarrier spacing, and can be divided into an integral part and a fractional part. The carrier phase, represented by $\theta(n)$, is a

Wiener process modelled by [1]:

$$\theta(n) - \theta(n - 1) = \nu(n). \quad (3)$$

The carrier phase is noisy or randomly time-varying, consisting of the sum of a constant term θ_0 and a sequence $\{\nu(n)\}_{n=0}^{N-1}$ of independent and identically distributed Gaussian random variables, each with mean zero and variance $\sigma_p^2 = 2\pi\Delta\nu T_s$. The constant term θ_0 is the CPO that is normally thought of as the common phase error, and the sequence $\{\nu(n)\}_{n=0}^{N-1}$ is the CPN.

In order to illustrate clearly our conceptually novel joint ML estimation approach, we have to somewhat limit the complexity of our system model. That way, it is analytically tractable, and leads to explicit theoretical results that provide a practical implementation. We first assume the channel is non-dispersive so that the channel impulse response is just an impulse, i.e., $h(n) = \delta(n)$. This assumption also means that the nonlinearities and dispersion that includes CD and PMD in the optical channel have been totally compensated and removed by optical means. This assumption has been commonly used in the recent literature [16], [21]–[23].

In addition, we assume in the estimator derivation that the CPN variance σ_p^2 is zero, and therefore the carrier phase is a constant and equal to the CPO θ_0 . This assumption is commonly used in practice [24], [25], because tracking a time-varying phase introduces much more analytical difficulties. Later, we will use simulations to show the estimation performance with the appearance of CPN. The theoretical development incorporating on the dispersive channel and the phase noise will be considered in our future work. Here we focus on the estimation of CFO, CPO and TO, which are all assumed to be constant. Then based on these assumptions, the signal model in (1) can be simplified to:

$$r(n) = x(n - \tau) e^{j(\frac{2\pi\epsilon n}{N} + \theta)} + w(n), n = 0, 1, \dots, N - 1. \quad (4)$$

In addition, in order to cope with the TO estimation range of $[0, N-1]$, we assume that the first two OFDM symbols are pilot symbols which are known to the receiver, and the second OFDM symbol is identical to the first. Assume without loss of generality that the pilot symbols are *MPSK* modulated, so that they have the constant amplitude, i.e., $|X(k)|^2 = A$.

The time-domain SNR γ_t is defined as:

$$\gamma_t = \frac{E_x}{\sigma^2}, \quad (5)$$

where E_x denotes the average energy per transmitted sample in the time domain, i.e., $E_x = E[|x(n)|^2]$. In Appendix A, we will show that the time-domain SNR is the same as the frequency-domain SNR, γ_f , defined as:

$$\gamma_f = \frac{E_s}{\sigma_W^2}, \quad (6)$$

where E_s denotes the average energy per subcarrier, i.e., $E_s = E[|X(k)|^2]$, where $k = 0, 1, \dots, N-1$. Term σ_W^2 is the variance of $W(k)$, where $\{W(k)\}_{k=0}^{N-1}$ is the frequency-domain AWGN that results from performing the N -point DFT of the time-domain AWGN, i.e., $\{w(n)\}_{n=0}^{N-1}$. We show in Appendix A that:

$$E_s = NE_x \text{ and } \sigma_W^2 = N\sigma^2. \quad (7)$$

Thus, we get:

$$\gamma_f = \gamma_t. \quad (8)$$

From here on, we denote the SNR as γ .

III. JOINT MAXIMUM LIKELIHOOD PARAMETER ESTIMATION

A. Overall General Approach

The optimum approach for estimating the CFO, CPO and TO is to design a joint ML estimator of all the parameters based on the observations of the OFDM samples over one symbol interval. This subsection will explain the general principle of the joint ML parameter estimator.

Since $\{w(n)\}_{n=0}^{N-1}$ is a sequence of independent, identically distributed, complex, AWGN, the likelihood function $p(\{r(n)\}_{n=0}^{N-1}|\tau, \epsilon, \theta)$ can be written as:

$$p(\{r(n)\}_{n=0}^{N-1}|\tau, \epsilon, \theta) = \left(\frac{1}{\pi\sigma^2}\right)^N \exp \left[-\sum_{n=0}^{N-1} \frac{|r(n) - x(n-\tau)e^{j[2\pi\epsilon n/N + \theta]}|^2}{\sigma^2} \right]. \quad (9)$$

The ML estimates $(\hat{\tau}, \hat{\epsilon}, \hat{\theta})$ of the parameters (τ, ϵ, θ) are the values which jointly maximize the likelihood function, i.e.:

$$(\hat{\tau}, \hat{\epsilon}, \hat{\theta}) = \arg \max_{\tau, \epsilon, \theta} p(\{r(n)\}_{n=0}^{N-1}|\tau, \epsilon, \theta). \quad (10)$$

Thus the ML estimates $(\hat{\tau}, \hat{\epsilon}, \hat{\theta})$ are the solutions of (11), which are shown as follows:

$$\begin{aligned} \frac{\partial p(\{r(n)\}_{n=0}^{N-1}|\tau, \epsilon, \theta)}{\partial \tau} &= 0, \\ \frac{\partial p(\{r(n)\}_{n=0}^{N-1}|\tau, \epsilon, \theta)}{\partial \epsilon} &= 0, \end{aligned}$$

$$\frac{\partial p(\{r(n)\}_{n=0}^{N-1}|\tau, \epsilon, \theta)}{\partial \theta} = 0. \quad (11)$$

It is obvious from the literature [24], [26] that it is analytically difficult even if not impossible to obtain explicit solutions of the three equations in (11). Thus, we propose here an alternative approach for joint estimation of CFO, CPO and TO, by using the communication theoretic concept of viewing the ML estimator as the minimum-error-probability signal detector. Supposing the signal samples after removing the CP given in (4) have been received. We pad $(L-1)N$ zeros at the end of the N received samples, so that we get a sequence of LN samples $\{r_{LN}(n)\}_{n=0}^{LN-1}$, which can be expressed as:

$$r_{LN}(n) = \begin{cases} r(n), & n = 0, 1, \dots, N-1 \\ 0, & n = N, N+1, \dots, LN-1 \end{cases}. \quad (12)$$

Here, we call L as the zero-padding factor, which is a power of 2. Then by taking the LN -point DFT of $r_{LN}(n)$, we obtain the noisy received OFDM symbol with CFO at LN discrete frequency points, i.e., $\{\frac{2\pi}{LN}k\}_{k=0}^{LN-1}$. The received sample on the k -th frequency $R_{LN}(k)$ in the frequency domain, for each $k = 0, 1, \dots, LN-1$, can be written as:

$$\begin{aligned} R_{LN}(k) &= \sum_{n=0}^{LN-1} r_{LN}(n)e^{-j2\pi kn/(LN)} \\ &= \sum_{n=0}^{N-1} x(n-\tau)e^{-j[\frac{2\pi n(k-\epsilon L)}{LN} - \theta]} + W_{LN}(k), \end{aligned} \quad (13)$$

where we have $W_{LN}(k) = \sum_{n=0}^{N-1} w(n)e^{-j\frac{2\pi kn}{LN}}$. Next, we define:

$$X_{LN}(k) = \sum_{n=0}^{N-1} x(n-\tau)e^{-j2\pi k(n-\tau)/(LN)}, \quad (14)$$

where $k = 0, 1, \dots, LN-1$. Actually, term $\{X_{LN}(k)\}_{k=0}^{LN-1}$ is the LN -point DFT of the zero-padded LN -point time-domain samples $\{\{x(n-\tau)\}_{n=0}^{N-1}, 0, \dots, 0\}$. Based on (13) and the DFT property that a time shift in the time domain causes a phase shift in the frequency domain, and vice versa, (13) can be written as:

$$\begin{aligned} R_{LN}(k) &= \left[\sum_{n=0}^{N-1} x(n-\tau)e^{-j\frac{2\pi(n-\tau)(k-\epsilon L)}{LN}} \right] \\ &\quad \cdot e^{-j[\frac{2\pi\tau(k-\epsilon L)}{LN} - \theta]} \\ &\quad + W_{LN}(k) \\ &= X_{LN}(k-\epsilon L)e^{-j[\frac{2\pi\tau(k-\epsilon L)}{LN} - \theta]} + W_{LN}(k). \end{aligned} \quad (15)$$

First, we think of the calculated spectrum of LN samples in (15) as being made up of L sets where the l -th set S_l is given by:

$$S_l = \{R_{LN}(k), k = l, l+L, \dots, l+(N-1)L\}. \quad (16)$$

This set S_l is the spectrum that corresponds to the hypothesized value $\tilde{\epsilon}_l$ for the CFO, which can be expressed as:

$$\tilde{\epsilon}_l = \frac{2\pi}{LN}l, \quad l = 0, 1, \dots, L-1. \quad (17)$$

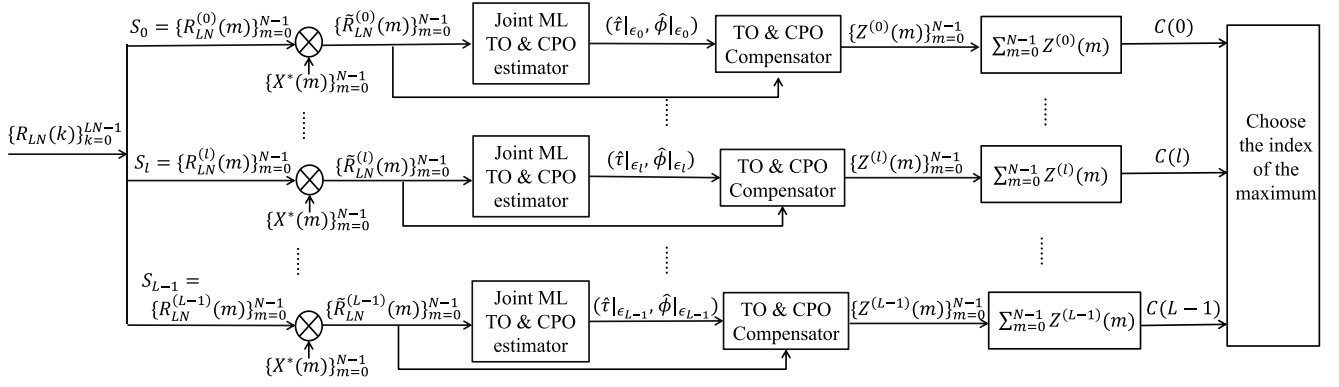


Fig. 2. Schematic diagram the joint MF ML parameter estimator, with the assumption that $\tau \in [0, N-1]$ and the first two OFDM symbols are known and identical to each other.

For simplicity, we first assume that L is sufficiently large so that the actual CFO is equal to one of the hypothesized values in $\{\tilde{\epsilon}_l\}_{l=0}^{L-1}$. The fine search will be introduced later. Note that for each set S_l , the set of noise samples, $W_l = \{W_{LN}(k), k = l, l+L, \dots, l+(N-1)L\}$, is a set of AWGN samples. It is easy to show that the elements in W_l are mutually independent of one another, each with mean 0 and variance $\sigma_W^2 = N\sigma^2$. However, considering all the elements in $\{W_{LN}(k)\}_{k=0}^{LN-1}$, the noise samples are not all independent of one another [27].

The hypothesized values of CFO in (17) are actually the possible radian frequencies of the CFO, thus the corresponding possible values of the normalized CFO are $\{\epsilon_l = l/L\}_{l=0}^{L-1}$, and we have $l = \epsilon_l L$. In the subset S_l given in (16), the m -th element at frequency point $k = l + mL$ is given by:

$$\begin{aligned} R_{LN}(k)|_{k=l+mL} &= R_{LN}(\epsilon_l L + mL) \\ &= X_{LN}((\epsilon_l - \epsilon)L + mL) e^{-j\left[\frac{2\pi(\epsilon_l - \epsilon + m)\tau}{N} - \theta\right]} + W_{LN}(\epsilon_l L + mL). \end{aligned} \quad (18)$$

Here, by defining

$$R_{LN}(k)|_{k=l+mL} \triangleq R_{LN}^{(l)}(m), \quad (19)$$

the l -th set S_l in (16) can be rewritten as:

$$S_l = \{R_{LN}^{(l)}(m), m = 0, 1, \dots, N-1\}. \quad (20)$$

For further simplification, we define:

$$V(m) = W_{LN}(\epsilon_l L + mL), m = 0, 1, \dots, N-1. \quad (21)$$

Since for each l , $\{W_{LN}(\epsilon_l L + mL)\}_{m=0}^{N-1}$ is actually an N -sample complex AWGN in the index m , with each element having mean 0 and variance $N\sigma^2$, and each sample can be written as $V(m) \sim \mathcal{CN}(0, N\sigma^2)$. Then, (18) can be further simplified as:

$$R_{LN}^{(l)}(m) = X_{LN}((\epsilon_l - \epsilon)L + mL) e^{-j\left[\frac{2\pi(\epsilon_l - \epsilon + m)\tau}{N} - \theta\right]} + V(m), \quad (22)$$

where we have $m = 0, 1, \dots, N-1$ and $l = 0, 1, \dots, L-1$. Equation (22) gives the model of the m -th element in the l -th set S_l in (16).

The schematic diagram of the joint ML parameter estimator is shown in Fig. 2. The received LN -point spectrum $\{R_{LN}(k)\}_{k=0}^{LN-1}$ is first demultiplexed into L sets, where the input to the l -th branch in the figure is the l -th set S_l , which corresponds to the hypothesized CFO ϵ_l . In the l -th branch, if the hypothesis that $\epsilon = \epsilon_l$ is correct, (22) can be expressed as:

$$R_{LN}^{(l)}(m)|_{\epsilon=\epsilon_l} = X_{LN}(mL) e^{-j\left[\frac{2\pi m\tau}{N} - \theta\right]} + V(m), \quad (23)$$

where based on (14), we have:

$$X_{LN}(mL) = \sum_{n=0}^{N-1} x(n-\tau) e^{-j2\pi m(n-\tau)/N} = X(m), \quad (24)$$

since the first two OFDM symbols are identical, the TO τ is a circular shift of $\{x(n)\}_{n=0}^{N-1}$. Term $X(m)$ is the DFT of the transmitted samples which is defined in (2). Thus, (23) can be further simplified to:

$$R_{LN}^{(l)}(m)|_{\epsilon=\epsilon_l} = X(m) e^{-j\left[\frac{2\pi m\tau}{N} - \theta\right]} + V(m). \quad (25)$$

The result in (25) shows that if the hypothesis $\epsilon = \epsilon_l$ is correct, then the l -th signal set $S_l = \{R_{LN}^{(l)}(m)\}_{m=0}^{N-1}$ would consist of the signal $\{X(m) e^{-j\left[\frac{2\pi m\tau}{N} - \theta\right]}\}_{m=0}^{N-1}$ plus AWGN $\{V(m)\}_{m=0}^{N-1}$. Since all the hypotheses $\{\epsilon = \epsilon_l\}_{l=0}^{L-1}$ are assumed equally likely, it is well known from communication theory that the most likely hypothesis is that $\epsilon = \epsilon_l$ for which the Euclidean distance between the l -th signal set S_l and the desired signal $\{X(m)\}_{m=0}^{N-1}$ is a minimum. This Euclidean distance is given by:

$$\begin{aligned} &\sum_{m=0}^{N-1} \left| R_{LN}^{(l)}(m) - X(m) \right|^2 \\ &= \sum_{m=0}^{N-1} \left| R_{LN}^{(l)}(m) \right|^2 + \sum_{m=0}^{N-1} |X(m)|^2 \\ &\quad - 2\Re \left\{ \sum_{m=0}^{N-1} R_{LN}^{(l)}(m) X^*(m) \right\}, \end{aligned} \quad (26)$$

where $\Re\{\cdot\}$ means taking the real part. We note that our decision problem here involves only one message, namely, $\{X(m)\}_{m=0}^{N-1}$, and L possible noisy received signals, namely,

$S_l, l = 0, \dots, L-1$. The problem is to determine the most likely received signal. In the common receiver decision problem, there is only one noisy received signal and multiple possible messages from which the receiver picks one as the decision. The conceptual difference between the two problems is small. Another point to note is that the Euclidean distance in (26) also depends on τ and θ , and has to be minimized with respect to these two parameter. As can be seen from (22) and (26), term $\sum_{m=0}^{N-1} |R_{LN}^{(l)}(m)|^2$ provides no information about ϵ . Thus, it can be discarded. Term $\sum_{m=0}^{N-1} |X(m)|^2$ is a constant because the data is modulated onto MPSK. Even for MQAM, this is a constant for a given OFDM spectrum, and contains no information about ϵ , thus can also be discarded. Therefore, the ML estimator chooses the hypothesis $\epsilon = \epsilon_l$ that maximizes the term:

$$C(l) = \Re \left\{ \sum_{m=0}^{N-1} R_{LN}^{(l)}(m) X^*(m) \right\}. \quad (27)$$

As shown in Fig. 2, the receiver generates the sequence $\{\tilde{R}_{LN}^{(l)}(m)\}_{m=0}^{N-1}$ for each signal set $S_l, l = 0, 1, \dots, N-1$, where

$$\tilde{R}_{LN}^{(l)}(m) = R_{LN}^{(l)}(m) X^*(m). \quad (28)$$

If the hypothesis that $\epsilon = \epsilon_l$ is correct, we then have from (23) that

$$\tilde{R}_{LN}^{(l)}(m) = |X(m)|^2 e^{-j(\frac{2\pi m\tau}{N} - \theta)} + V(m) X^*(m), \quad (29)$$

where the phase of $X(m)$ is removed. Since $X(m)$ is MPSK modulated, its amplitude is a constant, i.e., $|X(m)|^2 = A$. Thus (29) can be further written as:

$$\tilde{R}_{LN}^{(l)}(m) = A e^{-j(\frac{2\pi m\tau}{N} - \theta)} + W(m). \quad (30)$$

Here we have $W(m) = V(m) X^*(m)$, where $\{W(m)\}_{m=0}^{N-1}$ is complex AWGN with each $W(m) \sim \mathcal{CN}(0, AN\sigma^2)$. At this point, we consider minimizing the Euclidean distance in (26) with respect to τ and θ by maximizing $C(l)$ in (27).

In (30), by regarding the index m as the ‘time’ index, the signal $\{\tilde{R}_{LN}^{(l)}(m)\}_{m=0}^{N-1}$ for the l -th hypothesized value of CFO in (30) can be seen as the samples of a complex single sinusoid observed in AWGN, with the frequency of $-2\pi\tau/N$, phase θ and constant amplitude A . Thus, the joint estimation of τ and θ corresponding to the l -th hypothesized value of the CFO reduces to the estimation of the parameters of a single sinusoid in noise. This estimation can be realized by either the ‘time-domain’ approach in [19] or ‘frequency-domain’ approach in [20]. We will show the details of the approaches in the next subsection.

For the l -th hypothesized value of CFO, based on (30), we can obtain the joint ML estimates of TO and CPO, i.e., $(\hat{\tau}_{|\epsilon_l}, \hat{\theta}_{|\epsilon_l})$. Then $(\hat{\tau}_{|\epsilon_l}, \hat{\theta}_{|\epsilon_l})$ is used for compensating the corresponding set $\{\tilde{R}_{LN}^{(l)}(m)\}_{m=0}^{N-1}$. The m -th element of the compensated signal at l -th branch can be expressed as:

$$Z^{(l)}(m) = \Re \left\{ \tilde{R}_{LN}^{(l)}(m) e^{j\left[\frac{2\pi m\hat{\tau}_{|\epsilon_l}}{N} - \hat{\theta}_{|\epsilon_l}\right]} \right\}, \quad (31)$$

where we have $m = 0, 1, \dots, N-1$ and $l = 0, 1, \dots, L-1$. As can be seen from (30) and (31), if the hypothesized CFO ϵ_l is

equal to the exact value of the CFO, and if the estimates of TO and CPO are very accurate, i.e., $\epsilon_l = \epsilon$, $\hat{\tau}_{|\epsilon_l} = \tau$ and $\hat{\theta}_{|\epsilon_l} = \theta$, then $Z^{(l)}(m)$ is a constant observed in AWGN, i.e.:

$$Z^{(l)}(m) = A + \Re\{W(m)\} = A + W_R(m). \quad (32)$$

Term $W_R(m)$ is the real part of $W(m)$, which is real AWGN, i.e., $W_R(m) \sim \mathcal{N}(0, \frac{AN}{2}\sigma^2)$. Following (27), we take the summation of $Z^{(l)}(m)$ over m , then find the set of $\{\epsilon_l, \hat{\tau}_{|\epsilon_l}, \hat{\theta}_{|\epsilon_l}\}_{l=0}^{L-1}$ that gives the best match. The matched signal $C(l)$ can be written as:

$$C(l) = \sum_{m=0}^{N-1} Z^{(l)}(m). \quad (33)$$

In (33), when there is no AWGN, there will be a single peak with amplitude of AN at $l = \epsilon L$.

Note that (30)-(33) hold if and only if ϵ_l matches with ϵ . Otherwise, $X_{LN}[(\epsilon_l - \epsilon)L + mL]$ will not match with $X(m)$, so that we cannot get a constant phase, and the phase after the correlation in (29) is time-varying and unknown. Specifically, if ϵ_l is not equal to the actual CFO ϵ , the constellation phase removal step in (28) will become:

$$\begin{aligned} \tilde{R}_{LN}^{(l)}(m)|_{\epsilon_l \neq \epsilon} &= R_{LN}^{(l)}(m)|_{\epsilon_l \neq \epsilon} X^*(m) \\ &= X_{LN}[(\epsilon_l - \epsilon)L + mL] X^*(m) e^{-j[\frac{2\pi(\epsilon_l - \epsilon + m)\tau}{N} - \theta]} \\ &\quad + V(m) X^*(m) \\ &= \left[A^{(l)}(m) e^{-j[\frac{2\pi(\epsilon_l - \epsilon)\tau}{N}]} \right] e^{-j(\frac{2\pi m\tau}{N} - \theta)} + W(m), \end{aligned} \quad (34)$$

where we have

$$A^{(l)}(m) = X_{LN}[(\epsilon_l - \epsilon)L + mL] X^*(m). \quad (35)$$

The detailed derivation of the expression for $A^{(l)}(m)$ is shown in Appendix B. By comparing (34) and (30), it can be observed that the only difference is the amplitude. In (30), the amplitude is a real constant A , while in (34), i.e., in the case where ϵ_l is not equal to the actual CFO ϵ , the amplitude is $A^{(l)}(m) e^{-j[\frac{2\pi(\epsilon_l - \epsilon)\tau}{N}]}$ which is complex, depends on ϵ and τ , and varies with the ‘time-index’ m . To our knowledge, there is no existing technique to estimate τ and θ accurately for a single sinusoid with an unknown time-varying amplitude and phase. All existing methods, such as those in [19], [20], will generate very inaccurate estimates due to the phase disturbances caused by the mismatch. Thus, the joint ML parameter estimator works in favour of the correct set that corresponds to the actual CFO.

In addition, in (34), even if TO and CPO are perfectly compensated, the final correlator output signal is still smaller than in the case of $\epsilon_l = \epsilon$. Here, based on (31) and (B.4), and supposing there is no AWGN, the correlator output signal can be expressed as:

$$C(l)|_{\epsilon_l \neq \epsilon} = \Re\{AN e^{-j2\pi(\epsilon_l - \epsilon)/N}\} = AN \cos(2\pi(\epsilon_l - \epsilon)/N), \quad (36)$$

which is obviously less than AN .

Finally, the joint ML parameter estimates $(\hat{\epsilon}, \hat{\tau}, \hat{\theta})$ can be obtained by finding the index $\hat{l}$ which gives the best match in the

sense of maximizing $\{C(l)\}_{l=0}^{L-1}$, i.e.:

$$\hat{l} = \arg \max_l C(l). \quad (37)$$

Then we have:

$$\hat{\epsilon} = \epsilon_{\hat{l}}, \quad \hat{\tau} = \hat{\tau}_{|\epsilon_{\hat{l}}}, \quad \hat{\theta} = \hat{\theta}_{|\epsilon_{\hat{l}}}. \quad (38)$$

In the next subsection, we will show how to do the joint ML estimation of τ and θ for each hypothesized value of CFO.

B. Joint ML Estimation of TO and CPO

1) *'frequency-Domain' Approach:* To estimate the TO and the CPO, here we follow the approach in [20]. In the case of $\epsilon_l = \epsilon$, by regarding the index m as the 'time' index, and taking the N -point DFT of the 'time-domain' samples $\{\tilde{R}_{LN}^{(l)}(m)\}_{m=0}^{N-1}$ in (30), we introduce another frequency index ν , and compute the spectrum as:

$$\begin{aligned} R_F(\nu) &= \sum_{m=0}^{N-1} \tilde{R}_{LN}^{(l)}(m) e^{-j2\pi m\nu/N} \\ &= A \sum_{m=0}^{N-1} e^{-j[2\pi m(\tau+\nu)/N - \theta]} + W_F(\nu), \end{aligned} \quad (39)$$

where we have $\nu = 0, 1, \dots, N-1$ and $W_F(\nu) = \sum_{m=0}^{N-1} W(m) e^{-j2\pi m\nu/N}$.

As can be seen in (39), if ϵ_l is equal to ϵ , in the absence of noise, we can get a single peak of the amplitude spectrum at the frequency of $\nu = -\tau$, which is equal to AN . Thus, the ML estimates of the TO and CPO, i.e., $\hat{\tau}_F|_{\epsilon_l=\epsilon}$ and $\hat{\theta}_F|_{\epsilon_l=\epsilon}$, can be obtained by searching for the frequency index ν that maximizes $|R_F(\nu)|$, and the phase of the peak is the estimated CPO, i.e.,

$$\begin{aligned} \hat{\tau}_F|_{\epsilon_l=\epsilon} &= -\arg \max_{\nu} |R_F(\nu)|, \\ \hat{\theta}_F|_{\epsilon_l=\epsilon} &= \angle\{R_F(-\hat{\tau}_F|_{\epsilon_l=\epsilon})\}. \end{aligned} \quad (40)$$

Since τ is an integer, when SNR is low, the estimation error variance of τ may become large. To improve the estimation accuracy, as was done in [20], we pad $L'(N-1)$ zeros after $\{\tilde{R}_{LN}^{(l)}(m)\}_{m=0}^{N-1}$ where L' is the corresponding zero-padding factor, then perform the $L'N$ -point DFT as was done in (39). Similar to our MF ML CFO estimator, a fine search is required as shown later in (48). Reference [20] has indicated that the zero-padding rate of four for the coarse search is enough to get a good estimation performance.

However, in the case that the hypothesized CFO ϵ_l is not equal to the exact value of CFO, i.e., $\epsilon_l \neq \epsilon$, when performing

the N -point DFT as in (39), we have:

$$\begin{aligned} R_F(\nu)|_{\epsilon_l \neq \epsilon} &= \sum_{m=0}^{N-1} \tilde{R}_{LN}^{(l)}|_{\epsilon_l \neq \epsilon}(m) e^{-j2\pi m\nu/N} \\ &= e^{-j[\frac{2\pi(\epsilon_l - \epsilon)\tau}{N} - \theta]} \sum_{m=0}^{N-1} A^{(l)}(m) e^{-j\frac{2\pi m(\tau + \nu)}{N}} + W_F(\nu). \end{aligned} \quad (41)$$

By taking the absolute value of $R_F(\nu)|_{\epsilon_l \neq \epsilon}$, we can obtain from the detailed calculation of (41) in (B.4) that:

$$|R_F(\nu)|_{\epsilon_l \neq \epsilon} = \left| e^{-j[\frac{2\pi(\epsilon_l - \epsilon)\tau}{N} - \theta]} \cdot 0 + W_F(\nu) \right| = |W_F(\nu)|. \quad (42)$$

It can be observed from (41) that if $\epsilon_l \neq \epsilon$, we will obtain a broadened spectrum. Also, comparing (42) with (39) that has a single peak at the frequency $\nu = -\tau$, the amplitude $|R_F(\nu)|_{\epsilon_l \neq \epsilon}$ is just the amplitude of AWGN. Therefore, we can obtain the single peak if and only if $\epsilon_l = \epsilon$.

2) *'Time-Domain' Approach:* As mentioned above, the joint estimation of TO and CPO can also be realized in the 'time domain'. In [19], the frequency and phase of a complex single sinusoid with constant amplitude are obtained by solving the likelihood function. Explicit analytical results are given in [15, Eq. (16) and (17)]. Here, in the case of $\epsilon_l = \epsilon$, by applying our previous ML approach in [19], the estimation of TO and CPO reduces directly to calculating the frequency and phase of a sinusoid, respectively. The corresponding analytical results of $\hat{\tau}_T|_{\epsilon_l=\epsilon}$ and $\hat{\theta}_T|_{\epsilon_l=\epsilon}$ are expressed in (43) and (44), as shown at the bottom of this page. In addition, it is obvious that when $\epsilon_l \neq \epsilon$, using (43) and (44) will lead to very inaccurate results since $\tilde{R}_{LN}^{(l)}(m)|_{\epsilon_l \neq \epsilon}$ is a ϵ -dependent term, as can be seen from (34). As can be seen in (43) and (44), since the actual phase data sample is obtained from the principal argument of $\tilde{R}_{LN}^{(l)}(m)$, which is within the interval $[-\pi, \pi)$, phase unwrapping is required to generate $\angle \tilde{R}_{LN}^{(l)}(m)$. Here, we utilize the differential phase unwrapping method proposed in [28], to unwrap the phase $\angle \tilde{R}_{LN}^{(l)}(m)$ into the correct range. To unwrap the phase, we first calculate the phase difference between two adjacent samples, i.e., $\Delta(m) = \arg[\tilde{R}_{LN}^{(l)}(m+1)\tilde{R}_{LN}^{(l)}(m)^*]$. The first data sample at time $m=0$, is an observation on the phase ϕ alone. If $-\pi \leq \tilde{R}_{LN}^{(l)}(0) \leq \pi$, then a sequence of angle data $\{\psi(m)\}_{m=0}^{N-1}$ can be computed as:

$$\begin{aligned} \psi(1) &= \angle \tilde{R}_{LN}^{(l)}(0) + \Delta(0), \\ &\vdots \end{aligned}$$

$$\hat{\tau}_T|_{\epsilon_l=\epsilon} = \frac{N}{2\pi} \frac{\left[\sum_{m=0}^{N-1} m |\tilde{R}_{LN}^{(l)}(m)| \angle \tilde{R}_{LN}^{(l)}(m) \right] \left[\sum_{m=0}^{N-1} |\tilde{R}_{LN}^{(l)}(m)| \right] - \left[\sum_{m=0}^{N-1} |\tilde{R}_{LN}^{(l)}(m)| \angle \tilde{R}_{LN}^{(l)}(m) \right] \left[\sum_{m=0}^{N-1} m |\tilde{R}_{LN}^{(l)}(m)| \right]}{\left[\sum_{m=0}^{N-1} m^2 |\tilde{R}_{LN}^{(l)}(m)| \right] \left[\sum_{m=0}^{N-1} |\tilde{R}_{LN}^{(l)}(m)| \right] - \left[\sum_{m=0}^{N-1} m |\tilde{R}_{LN}^{(l)}(m)| \right]^2} \quad (43)$$

$$\hat{\theta}_T|_{\epsilon_l=\epsilon} = \frac{\left[\sum_{m=0}^{N-1} m |\tilde{R}_{LN}^{(l)}(m)| \angle \tilde{R}_{LN}^{(l)}(m) \right] \left[\sum_{m=0}^{N-1} m |\tilde{R}_{LN}^{(l)}(m)| \right] - \left[\sum_{m=0}^{N-1} m^2 |\tilde{R}_{LN}^{(l)}(m)| \right] \left[\sum_{m=0}^{N-1} |\tilde{R}_{LN}^{(l)}(m)| \angle \tilde{R}_{LN}^{(l)}(m) \right]}{\left[\sum_{m=0}^{N-1} m |\tilde{R}_{LN}^{(l)}(m)| \right]^2 - \left[\sum_{m=0}^{N-1} m^2 |\tilde{R}_{LN}^{(l)}(m)| \right] \left[\sum_{m=0}^{N-1} |\tilde{R}_{LN}^{(l)}(m)| \right]}. \quad (44)$$

$$\begin{aligned}
\psi(m+1) &= \psi(m) + \Delta(m) = \angle \tilde{R}_{LN}^{(l)}(0) + \Delta(0) \\
&+ \cdots + \Delta(m) \\
&\vdots \\
\psi(N-1) &= \psi(N-2) + \Delta(N-2) = \angle \tilde{R}_{LN}^{(l)}(0) + \Delta(0) \\
&+ \cdots + \Delta(N-1). \tag{45}
\end{aligned}$$

In order to obtain more accurate observation data, the phase of $\tilde{R}_{LN}^{(l)}(m)$ is collected by unwrapping $\arg\{\tilde{R}_{LN}^{(l)}(m)\}$ to within a 2π -interval centered around the computed value $\psi(m)$, i.e., the value of $\angle \tilde{R}_{LN}^{(l)}(m)$ chosen is the one lying in the interval $[\psi(m) - \pi, \psi(m) + \pi]$, where $m = 0, 1, \dots, N-1$. The necessary condition for a good performance of this phase unwrapping operation is that $\Delta(m)$ satisfies [28]:

$$-\pi \leq \Delta(m) \leq \pi. \tag{46}$$

3) *Discussion of the ‘Time-Domain’ and ‘Frequency-Domain’ Approaches:* The ‘frequency-domain’ approach is conceptually easy to implement, and the ML estimation of TO and CPO does not involve the phase unwrapping issues. However, a search is required, and to get accurate estimation result, zero-padding needs to be performed which enlarges the number of searches and increases the complexity. Moreover, the probability of obtaining a local optimum result increases with the decrease of SNR.

The ‘time-domain’ approach gives explicit analytical expressions of the TO and CPO ML estimates without exhaustive search, and has much lower complexity compared with the ‘frequency-domain’ method. However, the ‘time-domain’ approach needs to perform phase unwrapping, and it is worth noting that at low SNR, both ‘time-domain’ and ‘frequency-domain’ methods show estimation performance degradation.

C. Fine Search

Since we assume that the actual CFO is equal to one of the hypothesized values in (17), to get accurate estimation performance of CFO, TO and CPO, the zero-padding factor L has to be sufficiently large, which may lead to a high computational complexity. To ensure both the estimation accuracy and low complexity, here we propose a scheme that involves a one-step fine search of the CFO, followed by a recalculation of joint ML estimates of TO and CPO.

Based on (33), the fine search is realized by using a numerical method, i.e., the secant method [20]. The fine search problem becomes locating the zero point in $C'(l)$ with $C''(l) < 0$. Here term $C'(l)$ and $C''(l)$ represent the first order and second order differentiation of $C(l)$ given in (27), respectively. The gradient calculation of a discrete signal is defined in the same way as MATLAB does, i.e.:

$$C'(l) = \begin{cases} [C(l+1) - C(l-1)]/2, & l = 1, 2, \dots, N-2 \\ C(2) - C(1), & l = 0 \\ C(N-1) - C(N-2), & l = N-1. \end{cases} \tag{47}$$

If $C'(\hat{l}) > 0$, then the index of the actual peak, denoted by $\tilde{l}$, is on the right hand side of $\hat{l}$. On the other hand, if $C'(\hat{l}) < 0$, $\tilde{l}$ is on the left hand side. Therefore, the final estimated peak index of $C'(l)$ by using the secant method is given by [20]:

$$\tilde{l} = \begin{cases} \frac{C'(\hat{l})(\hat{l}+1) - C'(\hat{l}+1)\hat{l}}{C'(\hat{l}) - C'(\hat{l}+1)}, & \text{if } C'(\hat{l}) > 0 \\ \frac{C'(\hat{l})(\hat{l}-1) - C'(\hat{l}-1)\hat{l}}{C'(\hat{l}) - C'(\hat{l}-1)}, & \text{if } C'(\hat{l}) < 0 \end{cases}. \tag{48}$$

Then based on (48), the final ML estimate of CFO can be expressed as:

$$\hat{\epsilon} = \tilde{l}/L. \tag{49}$$

After obtaining $\hat{\epsilon}$, we first do the CFO compensation, then calculate $\hat{\tau}$ and $\hat{\theta}$ by either the ‘time-domain’ [19] or the ‘frequency-domain’ [20] approach.

Based on the above analysis, we can easily conclude that the proposed joint ML approach has full CFO estimation range, TO estimation range of $[0, N-1]$, and CPO estimation range of $[-\pi, \pi]$. For CFO estimation, to obtain the fractional part, we just take the value of l from 0 to $L-1$ with the step of 1. To obtain the integral part, then the value of l should be taken from 0 to $(L-1)N$, with the step of N .

IV. SEQUENTIAL MAXIMUM LIKELIHOOD PARAMETER ESTIMATION

Since for each hypothesized value of CFO, the joint MF ML parameter estimator needs to search for all the possible values of τ and θ , and the fine search step needs to be performed, the resulting implementation complexity will be high. Here we further develop a sequential approach which is suboptimal and proposed for arriving at a much simpler implementation. The schematic diagram of the sequential MF ML estimator is shown in Fig. 3. In our sequential approach, the ML CFO estimate is first obtained by taking the sum of the magnitudes of the received signal set $\{R_{LN}^{(l)}(m)\}_{m=0}^{N-1}$ in (22), one for each hypothesized value ϵ_l of ϵ , i.e.,

$$\begin{aligned}
C_S(l) &= \sum_{m=0}^{N-1} |R_{LN}^{(l)}(m)| \\
&= \sum_{m=0}^{N-1} |X_{LN}[(\epsilon_l - \epsilon)L + mL] e^{-j \left[\frac{2\pi(\epsilon_l - \epsilon + m)\tau}{N} - \theta \right]} + V(m)|. \tag{50}
\end{aligned}$$

Assume that the noise term $V(m)$ is small or negligible, and we have:

$$C_S(l) \approx \sum_{m=0}^{N-1} |X_{LN}[(\epsilon_l - \epsilon)L + mL]|. \tag{51}$$

As can be seen in (50) and (51), the absolute value operation can filter out the effect of phase caused by TO and CPO, and when $\epsilon_l = \epsilon$, $C_S(l)$ can reach the maximum without knowing the information of the transmitted spectrum. Thus, the CFO estimate can be finally obtained by finding the one which achieves the

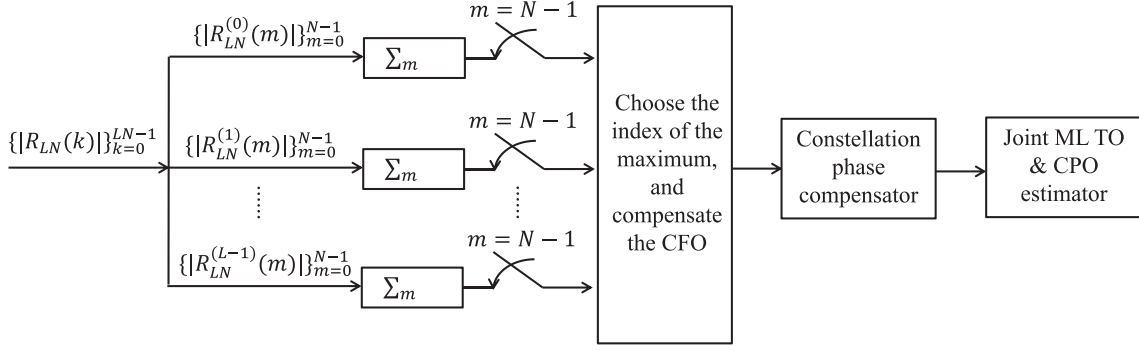


Fig. 3. Schematic diagram of the sequential MF ML estimator.

best match:

$$\hat{\epsilon}_S = \arg \max_{\epsilon_l} C_S(l). \quad (52)$$

Based on the estimated CFO $\hat{\epsilon}_S$ obtained in (52), we can use either the ‘time-domain’ approach [19] or the ‘frequency-domain’ approach [20] to obtain the ML estimates of TO and CPO jointly. One can see from the above development based on (50) and (51) that the sequential approach should work well at high SNR, and this is borne out by our simulation studies presented later in the paper. The sequential approach here forms the foundation of the approach we proposed earlier in [14].

In the sequential approach, since we utilize only the amplitude information of the spectrum for CFO estimation, the modulation format of MQAM will show a CFO estimation error variance worse than that of MPSK. This is because the MPSK modulated data symbols have the same amplitude, whereas the MQAM modulated data symbols have multi-level amplitudes, which may cause severe estimation error at low SNR. Therefore, the first two pilot OFDM symbols are modulated as MPSK format to ensure the CFO estimation performance.

In order to get accurate CFO estimation performance, the zero-padding factor needs to be large enough. To ensure a low complexity and a good estimation performance, similar to what has been done in Section III-C, a fine search needs to be performed as (48) and (49) show.

V. PERFORMANCE BOUNDS

To evaluate the estimation performance of the proposed algorithms, for the CFO estimation, we compared the proposed methods with Schmidl’s [8] method with respect to the inverse estimation error variance (IEEV). For the timing offset estimation, we compare the ‘time-domain’ and ‘frequency-domain’ algorithms with Schmidl’s [8], Minn’s [9], and Park’s [10] algorithms.

The estimation error variance is commonly used to measure the estimation accuracy. For ϵ , τ and θ , the corresponding estimation error variances are defined as:

$$\sigma_\epsilon^2 = E[(\hat{\epsilon} - \epsilon)^2], \sigma_\tau^2 = E[(\hat{\tau} - \tau)^2], \sigma_\theta^2 = E[(\hat{\theta} - \theta)^2]. \quad (53)$$

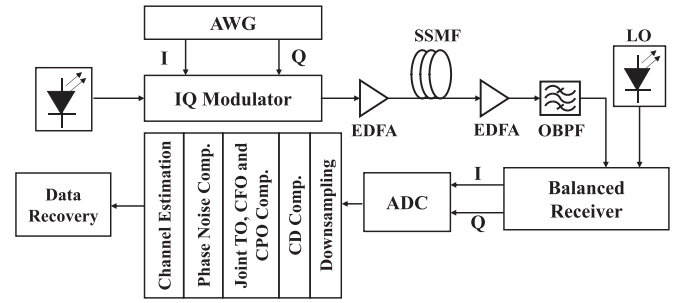


Fig. 4. Simulation setup of a CO-OFDM system. AWG: Arbitrary Waveform Generator; EDFA: Erbium-Doped Fiber Amplifier; SSMF: Standard Single Mode Fiber; OBPf: Optical Band-Pass Filter.

In Appendix C, we derive the following CRLBs on these variances:

$$\begin{aligned} \sigma_\epsilon^2 &\geq \frac{N^2(PN - S^2)}{8\pi^2 Q \gamma}, \sigma_\tau^2 \geq \frac{N^2(PN - S^2)}{8\pi^2 Q \gamma}, \\ \sigma_\theta^2 &\geq \frac{N^2 P^2 - S^4}{2QN^2 \gamma}, \end{aligned} \quad (54)$$

where the definition and expression of terms P , S and Q are shown in (C.6) and (C.8).

In addition, to evaluate the TO estimation performance, we also introduce the probability of correct timing estimation, which is computed as:

$$P_t = \frac{\text{The number of correct timing estimates}}{\text{Total number of timing estimates made}}. \quad (55)$$

The number of timing estimates made is 100000, to ensure reliable estimation of the probability.

VI. SIMULATION RESULTS AND DISCUSSION

A. System Setup

We refer to the transmitter and receiver block diagram of our CO-OFDM system shown in Fig. 1 and the setup in Fig. 4. The simulations are carried out through MATLAB and VPI TransmissionMaker 9.1. The length of serial input data bits is $2^{17} - 1$ and the number of data subcarriers is 256, which is the same as the DFT/IDFT size. The first two pilot symbols are modulated onto 16-PSK, while the payload data can be

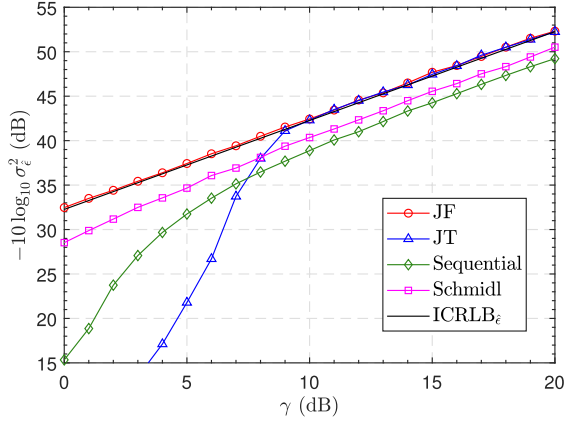


Fig. 5. IEEV of CFO vs. γ with $\epsilon = 0.3$, $\tau = 10$, and $\theta = \pi/4$.

modulated onto any modulation format. In our simulations, the payload data is under 16-QAM modulation. A cyclic prefix of 32 samples is added before each OFDM symbol and there are 128 OFDM symbols in total. The sample rate of the arbitrary waveform generator (AWG) is 25 GSa/s. In VPI simulations, the laser is working at 1550 nm, with the linewidth of 100 kHz and launch power of -2 dBm. After transmitting through the standard single mode fiber (SSMF) with the dispersion factor of 16 ps/nm/km, the signal is received by a balanced receiver and converted to digital by passing through the ADC with the sample rate of 50 GSa/s. In the digital signal processing procedure at the receiver side, we first downsample the received signal by using a median filter, and in our proposed MF ML algorithms, the zero-padding factor is selected to be 8 for CFO estimation, and 4 for the ‘frequency-domain’ approach for joint ML estimation of TO and CPO. For laser phase noise compensation in Fig. 4, we use the pilot-assisted algorithm [29], with 5 pilot subcarriers. Besides, there are 5 OFDM symbols used for channel estimation. Thus, the net data rate in the VPI simulation is $25 \text{ GSa/s} \times \log_2 16 \text{ bits/sample} \times \frac{121}{128} \times \frac{256}{288} \times \frac{251}{256} \approx 82.4 \text{ Gbps}$. Note that the results in subsections A through F are obtained using MATLAB simulations, while those in subsection G are obtained using VPI TransmissionMaker simulations.

To discuss the simulation results clearly, here we use abbreviations of the methods, defined as: JF/SF: joint/sequential ML estimator by the ‘frequency-domain’ approach, and JT/ST: joint/sequential ML estimator by the ‘time-domain’ approach. For the CFO estimation by the sequential method, we denote it as ‘sequential’ for simplicity.

B. Effects of SNR and CPN on CFO Estimation

Fig. 5 shows the CFO estimation performance of the proposed methods and Schmidl’s algorithm [8] versus SNR. The IEEV of the JT method deviates from the inverse CRLB (ICRLB) when SNR is smaller than 9 dB, while that of the JF method always reaches the ICRLB. At low SNR, the phase unwrapping error accumulates and leads to the degradation of the final estimation performance. For the sequential estimator, to get rid of the phase effect caused by TO, it utilizes the amplitude information only

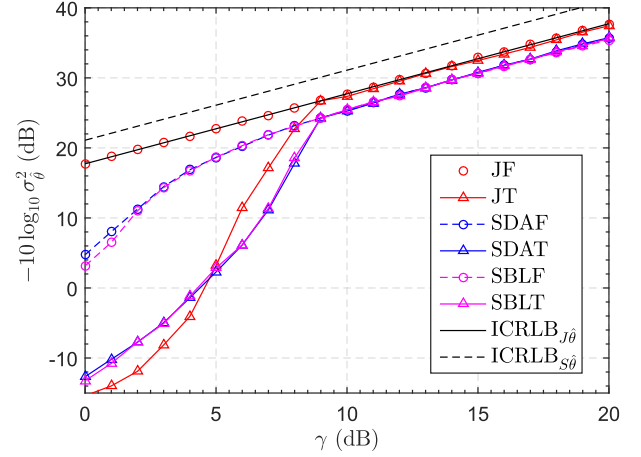


Fig. 6. IEEV of CFO vs. γ with $\epsilon = 0.3$, $\tau = 10$, and $\theta = \pi/4$ under the CLW of 50 kHz, 100 kHz and 200 kHz.

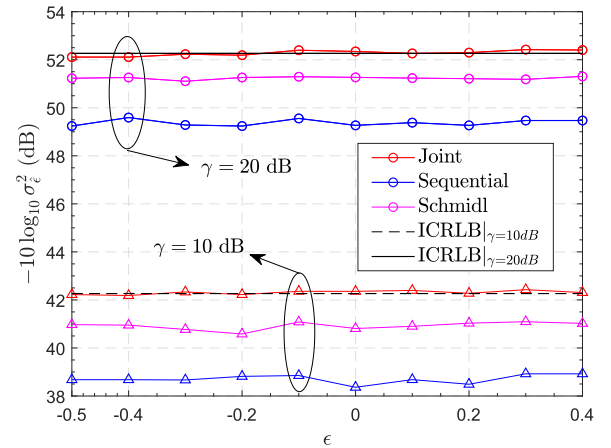


Fig. 7. IEEV vs. ϵ with under $\gamma = 10$ dB and $\gamma = 20$ dB.

for CFO estimation. Compared with the proposed joint ML estimator, the sequential method is suboptimal, thus its IEEV cannot reach the ICRLB.

In Fig. 6, in the presence of CPN, all the algorithms show performance degradation. However, the JF method always performs the best. The IEEV of the sequential method gradually converges to that of JF method with the increase of SNR. Under large CLW and high SNR, the sequential method outperforms Schmidl’s method. This is mainly because Schmidl’s method takes the correlation between the two identical halves in the time domain, and utilizes the phase information. Since the CPN is a Wiener process, the calculated phase after correlation actually includes the accumulated phase noise, which can lead to severe CFO estimation performance degradation at a high laser linewidth. However, the proposed methods perform well at the frequency domain, where the CPN after DFT, is composed of a common phase error (CPE) and an ICI part. When the CPN variance is small, the ICI part is commonly regarded as AWGN [1].

The IEEV performance versus different CFO values is shown in Fig. 7. The proposed methods have full CFO estimation range,

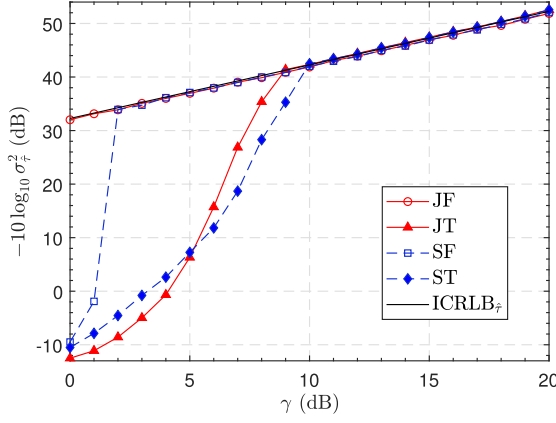


Fig. 8. IEEV of TO vs. γ with $\epsilon = 0.3$, $\tau = 10$, and $\theta = \pi/4$.

i.e., the integral part of CFO estimation range of $[0, N - 1]$, and the fractional part of CFO estimation range of $[-0.5, 0.5)$. Since Schmidl's method can only estimate the fractional part of CFO, here in Fig. 7, we only investigate the CFO range of $[-0.5, 0.5)$. It can be observed clearly that the IEEV of the proposed algorithms are transparent to the CFO values, and the performance of the joint method is the optimum, which is consistent with Fig. 5.

C. Effects of SNR on to Estimation

Fig. 8 shows the IEEV performance of the proposed methods versus SNR. The IEEV performance of TO of the JF and SF methods can achieve the ICRLB when SNR is higher than 2 dB. The SF method performs worse at lower SNR, because the CFO estimation error affects the estimation accuracy of TO, which is consistent with Fig. 5. The proposed JT and ST methods can perform well when SNR is higher than 10 dB, but due to the error caused by the phase unwrapping algorithm at low SNR, their IEEV degrades severely.

D. Timing Metrics

The timing metrics of the proposed JF/SF, Schmidl's, Minn's and Park's methods are shown in Fig. 9, with $\tau = 10$, under the case of CFO and CPO totally compensated. As can be seen from Fig. 9, the timing metric of Schmidl's method has a plateau that is equal to the length of CP, which causes severe timing estimation error at low SNR. Minn's and Park's methods solve this problem, and Park's method has a sharper peak than Minn's method. However, both Minn's and Park's methods have large side lobes which may cause severe timing estimation error because of leading to the wrong peak at low SNR. Compared with these algorithms, the timing metric of the proposed JF/SF method shows a clear and sharp peak without any side lobes.

E. Probability of Correct Timing Estimation

The timing estimation performance comparison of the proposed and conventional methods is shown in Fig. 10, with respect to the probability of correct timing estimation. In this simulation,

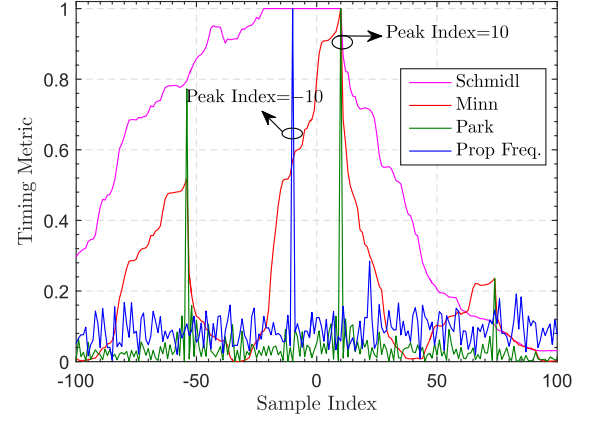


Fig. 9. Timing metrics of JF/SF, Schmidl's, Minn's and Park's methods, with $\tau = 10$, $\epsilon = 0$ and $\theta = 0$.

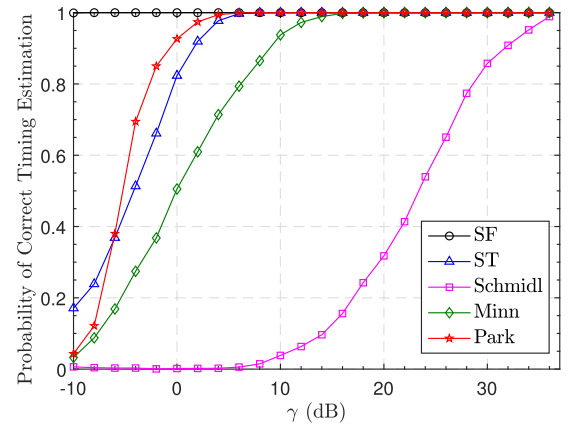


Fig. 10. Probability of correct timing estimation, with $\tau = 10$.

we assume the CFO has been totally compensated. For the SF and ST methods, we choose the nearest integer of the TO estimates as the estimation. In Fig. 10, Schmidl's method shows the worst timing estimation performance, because the plateau in the timing metric leads to its sensitivity to the AWGN, and causes severe performance degradation. In addition, since the timing metric of Park's method has a sharper main peak than that of Minn's method, the probability of correct timing estimation of Park's method is higher than that of Minn's method, especially at low SNR.

The SF method shows the best timing estimation performance, with the timing estimation accuracy of 1 at even -10 dB SNR. The ST method performs a little worse than Park's method at low SNR. There are two main reasons. First, the phase unwrapping algorithm is sensitive to the AWGN. Second, the TO estimate is obtained from the analytical result shown in (43), in which the TO related radian frequency is calculated first, then multiply by $N/2\pi$ to get the final estimate. The multiplication step enlarges the small error and causes fluctuations around the actual TO, thus decreasing the probability of timing estimation.

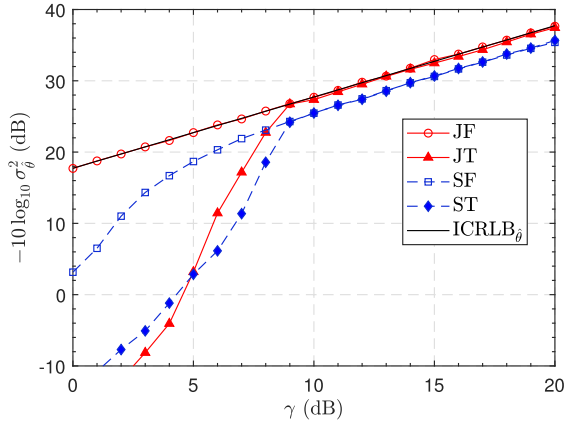


Fig. 11. IEEV of CPO vs. γ with $\epsilon = 0.3$, $\tau = 10$, and $\theta = \pi/4$.

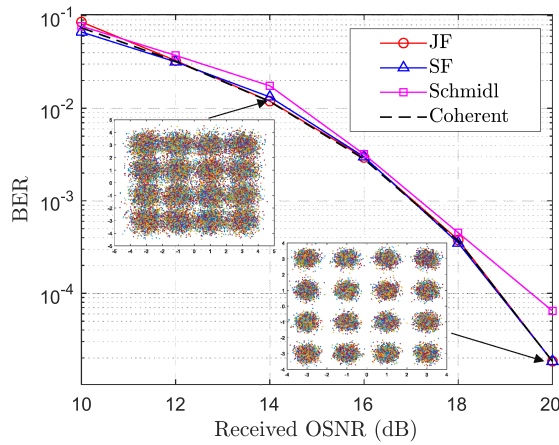


Fig. 12. BER vs. OSNR in back to back transmission, with the CFO of 30 MHz and the CLW of 200 kHz.

F. Effect of SNR on CPO Estimation

Fig. 11 shows the IEEV of the proposed methods vs. SNR. The IEEV of the JF method can always reach the ICRLB, while that of the JT method reaches the ICRLB when SNR is higher than 9 dB. At low SNR, the JT method shows performance degradation due to the error caused by the phase unwrapping algorithm. The IEEV of the sequential method cannot reach the ICRLB since the CFO estimation step is suboptimal, whose accuracy will affect the estimation of TO and CPO. The result in Fig. 11 is consistent with that of Fig. 5.

G. BER Performance and Nonlinearities Tolerance

The final BER performance of the proposed algorithms and Schmidl's method versus OSNR is investigated by using VPI TransmissionMaker, which is shown in Fig. 12, with the CFO of 30 MHz and the combined laser linewidth of 200 kHz, in a back to back transmission scenario. From Section VI. D, we can observe that the timing metric plateau of Schmidl's method can lead to severe degradation on the timing estimation accuracy in a noisy and dispersive channel, thus causing the degradation of BER performance. Therefore, in this simulation, we correct the

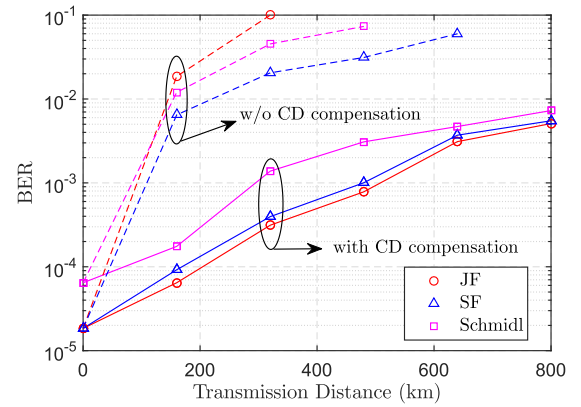


Fig. 13. BER vs. various transmission distances, with the CFO of 30 MHz and the CLW of 200 kHz, at the OSNR of 20 dB.

timing estimation error of Schmidl's method, in order to show a clear performance comparison of the CFO estimation. It can be observed from Fig. 12 that the BER performance of the proposed algorithms can always converge to the coherent bound, which is obtained by directly using the actual CFO value for compensation. Schmidl's algorithm performs a little worse, but the BER difference between these algorithms is not obvious. This is because the residual CFO can be compensated and corrected by the phase noise compensation algorithm that follows the joint MF ML estimation and compensation algorithm, as shown in Fig. 4.

We also investigate the CD and fiber nonlinearities tolerance of the proposed algorithms. Fig. 13 shows the BER performance after transmitting through various fiber lengths with the CFO of 30 MHz and the CLW of 200 kHz, at the OSNR of 20 dB. It can be clearly observed that with CD compensation [30], the proposed JF method shows the best performance, while Schmidl's method shows the worst. However, without the CD compensation, after a 100 km transmission, the BER of JF method is the highest, while the SF method is the lowest. This is because the data-aided algorithms such as JF and Schmidl's methods depend highly on the transmitted data. With the appearance of fiber nonlinearities, the pilot symbols used for CFO estimation have already been severely distorted, thus leading to a poor BER performance. However, the proposed SF method estimates the CFO, TO and CPO in a sequential manner, in which the CFO estimation is achieved in a blind manner by utilizing only the amplitude information, and thus it shows the best tolerance of CD and fiber nonlinearities. Therefore, we can easily conclude that the proposed MF ML algorithms are applicable to real long-haul optical fiber transmissions.

In order to investigate the performance of proposed algorithms in detail, we further plot the Q factor versus launch power, which is shown in Fig. 14. It can be seen that for 480 km transmission, the optimum launch power of the proposed algorithms is -1 dBm, while that of Schmidl's method is 1 dBm. This illustrates that Schmidl's method performs better under highly nonlinear transmission than JF and SF methods. The performance of proposed algorithms is worse than Schmidl's method when the

TABLE I
COMPLEXITY COMPARISON

Algorithms		Complex Mul.	Complex Add.	Typical Val.
Joint	Freq.	$\frac{LN}{2} \log_2(LN) + L'N \log_2(L'N) + 2N + 9$	$LN \log_2(LN) + 2L'N \log_2(L'N) + 2N + 8$	22025/43528
	Time	$\frac{LN}{2} \log_2(LN) + 16N + 23$	$LN \log_2(LN) + 16N - 2$	15383/26622
Seq.	Freq.	$\frac{LN}{2} \log_2(LN) + \frac{L'N}{2} \log_2(L'N) + N + 6$	$LN \log_2(LN) + L'N \log_2(L'N) + 5$	16646/ 32773
	Time	$\frac{LN}{2} \log_2(LN) + 8N + 13$	$LN \log_2(LN) + 7N$	13325/24320
Schmidl	Total	$(N^2 + 2N) + (\frac{N}{2} + 1)$	$(N^2 - 2N) + (\frac{N}{2} - 1)$	66177/65151

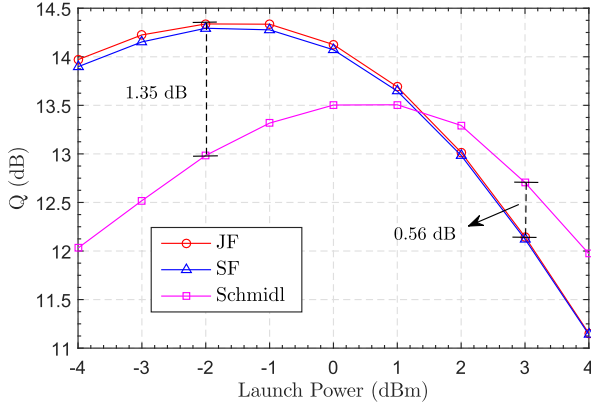


Fig. 14. Q factor versus launch power after 480 km transmission, with CD compensation.

power is above 1.2 dBm. However, when the Q factor is below 13.5 dB, the corresponding BER is above $1\text{E}-3$, which will become even worse at a higher power level. When the launch power is lower than 1.2 dBm, the proposed algorithms show obvious performance advantage compared with Schmidl's method. Therefore, although the performance of proposed algorithms degrades in highly nonlinear environments, in which case the BER is relatively high, the proposed algorithms show superior performance than Schmidl's method under the condition of lower nonlinearity. In a real transmission, by using existing algorithms to compensate for the dispersion and nonlinearities, the proposed algorithms are able to tolerate the residual effect, which can be seen from the simulation results in Fig. 13 and Fig. 14. Thus, the proposed algorithms show the possibility for applying to higher speed systems.

H. Complexity Comparison

Table I shows the complexity comparison between the proposed and Schmidl's methods. Some abbreviations are defined as: Freq./Time: frequency/time-domain approach, Mul.: multiplications, Add.: additions, Val.: values, Seq.: sequential method. Term L' is the zero-padding factor of JF/SF method. For the typical values in simulations, we have $L = 8$, $N = 256$ and $L' = 4$.

In Table I, the complexity of Schmidl's method is the highest. This is because the timing estimation of Schmidl's method requires searching for all the possible values, and calculating the timing metric within each search. The sequential method is simpler than the joint method, and the time-domain approach shows

lower complexity than the frequency-domain method. Because the time-domain method utilizes the analytical results while the frequency-domain method needs to perform the N -point FFT and a fine search. We should point out that our proposed methods not only have lower complexity, but also can achieve the joint estimation of CFO, TO and CPO, while Schmidl's method can only estimate the TO and CFO.

VII. CONCLUSION

Based on the concept of optimum signal detection in AWGN, we developed a joint ML estimator of the CFO, CPO and TO in the frequency domain. The resulting receiver is straightforward to implement, although it can be computationally heavy. However, the theoretical analysis demonstrates how one can derive a suboptimum sequential estimator that first obtains an approximate ML estimate of the CFO, followed by joint ML estimation of the CPO and TO. The sequential estimator is much simpler computationally, and its performance loss is small compared to the performance of the optimum joint estimator at high SNR. Also, simulation results show that the estimation performance of the sequential approach is more robust to the CD and the fiber nonlinearity. One should note the conceptual novelty of our approach here that is easy to understand and simple to implement compared to other existing approaches. The system model we picked here for our analysis may be simpler than that in other papers, for instance, [31]–[34], but we did this in order to make it easier for the reader to understand the principle of our approach. In a future report, we will extend our work to the case of channels with time and frequency dispersion.

APPENDIX A

DERIVATION OF THE RELATIONSHIP BETWEEN TIME-DOMAIN SNR AND FREQUENCY-DOMAIN SNR

Since the $\{w(n)\}_{n=0}^{N-1}$ is a sequence of independent, identically distributed, complex, AWGN random variables, then we can easily obtain that $\sigma_w^2 = N\sigma^2$. The average energy per subcarrier E_X can be rewritten as:

$$\begin{aligned}
 E_s &= E[|X(k)|^2] = E \left[\left| \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \right|^2 \right] \\
 &= E \left[\sum_{n=0}^{N-1} |x(n)|^2 \right]
 \end{aligned}$$

$$+ \Re \left\{ E \left[\sum_{n=0}^{N-1} \sum_{n_1=0, n_1 \neq n}^{N-1} x(n) x^*(n_1) e^{-j \frac{2\pi k(n-n_1)}{N}} \right] \right\}. \quad (\text{A.1})$$

Since the data on the subcarriers are independent with each other, and for M -PSK and M -QAM modulation, the constellation points are symmetric with equal probability, then we have:

$$E[X(k)] = 0, E[X(k)X^*(k_1)] = 0 \text{ (for } k \neq k_1\text{)}. \quad (\text{A.2})$$

Thus, after the N -point IDFT, we can obtain:

$$E[x(n)] = 0, E[x(n)x^*(n_1)] = 0 \text{ (for } n \neq n_1\text{)}. \quad (\text{A.3})$$

Then (A.1) can be simplified as:

$$E_s = E[|X(k)|^2] = E \left[\sum_{n=0}^{N-1} |x(n)|^2 \right] = NE_x. \quad (\text{A.4})$$

Finally, the frequency-domain SNR γ_f is:

$$\gamma_f = \frac{E_s}{\sigma_W^2} = \frac{NE_x}{N\sigma^2} = \gamma. \quad (\text{A.5})$$

APPENDIX B EXPRESSION OF $A^l(m)$

From (35), we can calculate $A^l(m)$ as:

$$\begin{aligned} A^l(m) &= X_{LN}[(\epsilon_l - \epsilon)L + mL]X^*(m) \\ &= \sum_{n=0}^{N-1} x(n - \tau) e^{-j \frac{2\pi(n-\tau)(\epsilon_l - \epsilon + m)}{N}} \sum_{n_1=0}^{N-1} x^*(n_1) e^{j \frac{2\pi n_1 m}{N}} \\ &= \sum_{n'=-\tau}^{N-1-\tau} x(n') e^{-j \frac{2\pi n'(\epsilon_l - \epsilon + m)}{N}} \sum_{n_1=0}^{N-1} x^*(n_1) e^{j \frac{2\pi n_1 m}{N}}. \end{aligned} \quad (\text{B.1})$$

According to the cycle characteristics of DFT, (B.1) can be further calculated as:

$$\begin{aligned} A^l(m) &= \sum_{n=n_1=0}^{N-1} |x(n)|^2 e^{-j \frac{2\pi(\epsilon_l - \epsilon)}{N}} \\ &+ \sum_{n=0}^{N-1} e^{-j \frac{2\pi(\epsilon_l - \epsilon)n}{N}} \sum_{n_1=0, n_1 \neq n}^{N-1} x(n)x^*(n_1) e^{-j \frac{2\pi m(n-n_1)}{N}}. \end{aligned} \quad (\text{B.2})$$

As can be seen from Appendix A, $\sum_{n=0}^{N-1} |x(n)|^2 = NE_x$, where we have $NE_x = E_s$ and for MPSK modulation, $E_s = A$. Thus, (B.2) can be simplified to:

$$\begin{aligned} A^l(m) &= A e^{-j \frac{2\pi(\epsilon_l - \epsilon)}{N}} \\ &+ \sum_{n=0}^{N-1} e^{-j \frac{2\pi(\epsilon_l - \epsilon)n}{N}} \sum_{n_1=0, n_1 \neq n}^{N-1} x(n)x^*(n_1) e^{-j \frac{2\pi m(n-n_1)}{N}}. \end{aligned} \quad (\text{B.3})$$

In Section III-B, we take the N -point DFT of $A^l(m) e^{-j 2\pi m \tau / N}$, which can be calculated as:

$$\begin{aligned} F_A^{(l)}(\nu) &= \sum_{m=0}^{N-1} A^l(m) e^{-j 2\pi m(\tau + \nu)/N} \\ &= A e^{-j \frac{2\pi(\epsilon_l - \epsilon)}{N}} \sum_{m=0}^{N-1} e^{-j \frac{2\pi m(\tau + \nu)}{N}} \\ &+ \sum_{n=0}^{N-1} e^{-j \frac{2\pi(\epsilon_l - \epsilon)n}{N}} \sum_{n_1=0, n_1 \neq n}^{N-1} x(n)x^*(n_1) \sum_{m=0}^{N-1} e^{-j \frac{2\pi m(n-n_1 + \tau + \nu)}{N}} \\ &= 0. \end{aligned} \quad (\text{B.4})$$

The third step of (B.4) is obtained based on the property of geometric progression [35], i.e., $\sum_{m=0}^{N-1} e^{-j \frac{2\pi m(\tau + \nu)}{N}} = \frac{1 - e^{-j 2\pi(\tau + \nu)}}{1 - e^{-j 2\pi(\tau + \nu)/N}} = 0$, and similarly, $\sum_{m=0}^{N-1} e^{-j \frac{2\pi m(\tau + \nu + n - n_1)}{N}} = 0$. This is because the terms $\tau + \nu$ and $\tau + \nu + n - n_1$ in the numerators are integers, then $e^{-j 2\pi(\tau + \nu)}$ and $e^{-j 2\pi(\tau + \nu + n - n_1)}$ are both equal to 1.

APPENDIX C

DERIVATION OF THE UNBIASED CRAMÉR-RAO LOWER BOUND

Here, $(\hat{\epsilon}, \hat{\tau}, \hat{\theta})$ become asymptotically unbiased as SNR increases, that have been proved in [14], [20]. The unbiased CRLBs are the diagonal elements of the inverse of the Fisher information matrix J , whose typical element is given by [20]:

$$J_{ij} = -E\{H_{z_i z_j}\}, \quad (\text{C.1})$$

$$J = \gamma \begin{bmatrix} \frac{8\pi^2}{N^2} P & -\frac{8\pi^2}{N^3} S^2 & \frac{4\pi}{N} S \\ -\frac{8\pi^2}{N^3} S^2 & \frac{8\pi^2}{N^2} P & -\frac{4\pi}{N} S \\ \frac{4\pi}{N} S & -\frac{4\pi}{N} S & 2N \end{bmatrix}, \quad (\text{C.5})$$

$$P = \sum_{n=0}^{N-1} n^2 = \frac{1}{6} N(N-1)(2N-1), \quad S = \sum_{n=0}^{N-1} n = \frac{1}{2} (N-1)N. \quad (\text{C.6})$$

$$J^{-1} = \frac{J^*}{|J|} = \frac{N^4}{\gamma 128\pi^4 Q} \begin{bmatrix} \frac{16\pi^2}{N} P - \frac{16\pi^2}{N^2} S^2 & 0 & \frac{32\pi^3}{N^4} S^3 - \frac{32\pi^3}{N^3} SP \\ 0 & \frac{16\pi^2}{N} P - \frac{16\pi^2}{N^2} S^2 & \frac{32\pi^3}{N^3} PS - \frac{32\pi^3}{N^4} S^3 \\ \frac{32\pi^3}{N^4} S^3 - \frac{32\pi^3}{N^3} SP & \frac{32\pi^3}{N^3} PS - \frac{32\pi^3}{N^4} S^3 & \frac{64\pi^4}{N^4} P^2 - \frac{64\pi^4}{N^6} S^4 \end{bmatrix}, \quad (\text{C.7})$$

$$Q = NP^2 + S^4/N - 2S^2P. \quad (\text{C.8})$$

where we have $\mathbf{z} = [\epsilon, \tau, \theta]^T$, and the expectation is with respect to the likelihood function, i.e.:

$$H_{z_i} = \frac{\partial}{\partial z_i} \log p(\{r(n)\}_{n=0}^{N-1} | \tau, \epsilon, \theta), \quad (\text{C.2})$$

where we have

$$\begin{aligned} & \log p(\{r(n)\}_{n=0}^{N-1} | \tau, \epsilon, \theta) \\ &= -N \log(\pi\sigma^2) - \frac{1}{\sigma^2} \sum_{n=0}^{N-1} \{|r(n)|^2 + |x(n - \tau)|^2\} \\ &+ \frac{2}{\sigma^2} \Re \left\{ \sum_{n=0}^{N-1} r(n) x^*(n - \tau) e^{-j[2\pi\epsilon n/N + \theta]} \right\}. \end{aligned} \quad (\text{C.3})$$

After calculation, the Fisher information matrix J can be obtained which is in (C.5), as shown at the bottom of the previous page. Then the inverse matrix J^{-1} is expressed in (C.7), also shown at the bottom of the previous page. Finally, then bounds are given by

$$\begin{aligned} \sigma_\epsilon^2 &\geq J_{11}^{-1} = \frac{N^2(PN - S^2)}{8\pi^2 Q \gamma}, \\ \sigma_\tau^2 &\geq J_{22}^{-1} = \frac{N^2(PN - S^2)}{8\pi^2 Q \gamma}, \\ \sigma_\theta^2 &\geq J_{33}^{-1} = \frac{N^2 P^2 - S^4}{2QN^2 \gamma} \end{aligned} \quad (\text{C.4})$$

where the expressions of P, S and Q are shown in (C.6) and (C.8).

REFERENCES

- [1] W. Shieh and I. Djordjevic, *OFDM for Optical Communications*. New York, NY, USA: Academic, 2009.
- [2] A. J. Coulson, "Maximum likelihood synchronization for OFDM using a pilot symbol: Algorithms," *IEEE J. Sel. Areas Commun.*, vol. 19, no. 12, pp. 2486–2494, Dec. 2001.
- [3] D. D. Lin, R. A. Pacheco, T. J. Lim, and D. Hatzinakos, "Joint estimation of channel response, frequency offset, and phase noise in OFDM," *IEEE Trans. Signal Process.*, vol. 54, no. 9, pp. 3542–3554, Sep. 2006.
- [4] J.-J. Van de Beek, M. Sandell, and P. O. Borjesson, "ML estimation of time and frequency offset in OFDM systems," *IEEE Trans. Signal Process.*, vol. 45, no. 7, pp. 1800–1805, Jul. 1997.
- [5] M. M. U. Gul, X. Ma, and S. Lee, "Timing and frequency synchronization for OFDM downlink transmissions using Zadoff–Chu sequences," *IEEE Trans. Wireless Commun.*, vol. 14, no. 3, pp. 1716–1729, Mar. 2015.
- [6] Y. Cao, W. Su, and S. N. Batalama, "A novel receiver design and maximum-likelihood detection for distributed MIMO systems in presence of distributed frequency offsets and timing offsets," *IEEE Trans. Signal Process.*, vol. 66, no. 23, pp. 6297–6309, Dec. 2018.
- [7] G. Miriyala, A. Kakumanu, M. Swapna, and P. S. Shankar, "Joint estimation of channel response, frequency offset and phase noise in OFDM systems," in *Proc. IEEE 2nd Int. Conf. Intell. Comput. Control Syst.*, 2018, pp. 1299–1303.
- [8] T. M. Schmidl and D. C. Cox, "Robust frequency and timing synchronization for ofdm," *IEEE Trans. Commun.*, vol. 45, no. 12, pp. 1613–1621, Dec. 1997.
- [9] H. Minn, M. Zeng, and V. K. Bhargava, "On timing offset estimation for OFDM systems," *IEEE Commun. Lett.*, vol. 4, no. 7, pp. 242–244, Jul. 2000.
- [10] B. Park, H. Cheon, C. Kang, and D. Hong, "A novel timing estimation method for OFDM systems," *IEEE Commun. Lett.*, vol. 7, no. 5, pp. 239–241, May 2003.
- [11] X. Zhou, K. Long, R. Li, X. Yang, and Z. Zhang, "A simple and efficient frequency offset estimation algorithm for high-speed coherent optical OFDM systems," *Opt. Express*, vol. 20, no. 7, pp. 7350–7361, 2012.
- [12] J. Wu *et al.*, "A robust and efficient frequency offset correction algorithm with experimental verification for coherent optical OFDM system," *J. Lightw. Technol.*, vol. 33, no. 18, pp. 3801–3807, 2015.
- [13] H.-G. Jeon, K.-S. Kim, and E. Serpedin, "An efficient blind deterministic frequency offset estimator for OFDM systems," *IEEE Trans. Commun.*, vol. 59, no. 4, pp. 1133–1141, Apr. 2011.
- [14] X. Du, T. Song, and P.-Y. Kam, "Carrier frequency offset for CO-OFDM: The matched filter approach," *J. Lightw. Technol.*, vol. 36, no. 14, pp. 2955–2965, 2018.
- [15] X. Du, J. Zhang, Y. Li, C. Yu, and P.-Y. Kam, "Efficient joint timing and frequency synchronization algorithm for coherent optical OFDM systems," *Opt. Express*, vol. 24, no. 17, pp. 19 969–19 977, 2016.
- [16] B. You *et al.*, "Joint carrier frequency offset and phase noise estimation based on pseudo-pilot in CO-FBMC/OQAM system," *IEEE Photon. J.*, vol. 11, no. 1, pp. 1–11, Feb. 2019.
- [17] K. Ren *et al.*, "A time and frequency synchronization method for CO-OFDM based on CMA equalizers," *Opt. Commun.*, vol. 416, pp. 166–171, 2018.
- [18] X. Du, P.-Y. Kam, and C. Yu, "Joint timing and frequency synchronization in coherent optical OFDM systems," *Frontiers Optoelectron.*, vol. 12, no. 1, pp. 4–14, 2019.
- [19] H. Fu and P. Y. Kam, "MAP/ML estimation of the frequency and phase of a single sinusoid in noise," *IEEE Trans. Signal Process.*, vol. 55, no. 3, pp. 834–845, Mar. 2007.
- [20] D. Rife and R. Boorstyn, "Single tone parameter estimation from discrete-time observations," *IEEE Trans. Inf. Theory*, vol. IT-20, no. 5, pp. 591–598, Sep. 1974.
- [21] L. Li, W. Yao, L. Han, and G.-J. Hu, "Timing synchronization algorithm based on FH sequence for coherent optical OFDM systems with carrier frequency offset," *Optoelectron. Lett.*, vol. 15, no. 4, pp. 288–291, 2019.
- [22] X. Fang, Y. Wang, D. Ding, and L. Zhang, "Pilot-aided phase noise suppression for coherent optical ofdm/oqam," in *Proc. Asia Commun. Photon. Conf.*, 2019, paper M4A.107.
- [23] J. Liu, Y. Wei, X. Zeng, J. Lu, S. Zhang, and M. Wang, "A novel joint timing/frequency synchronization scheme based on Radon–Wigner transform of LFM signals in CO-OFDM systems," *Opt. Commun.*, vol. 410, pp. 744–750, 2018.
- [24] H. Abdzadeh-Ziabari, W.-P. Zhu, and M. Swamy, "Joint maximum likelihood timing, frequency offset, and doubly selective channel estimation for OFDM systems," *IEEE Trans. Veh. Technol.*, vol. 67, no. 3, pp. 2787–2791, Mar. 2018.
- [25] H. Rahbari, M. Krunz, and L. Lazos, "Swift jamming attack on frequency offset estimation: The achilles' heel of OFDM systems," *IEEE Trans. Mobile Comput.*, vol. 15, no. 5, pp. 1264–1278, May 2016.
- [26] O. H. Salim, A. A. Nasir, H. Mehrpouyan, W. Xiang, S. Durrani, and R. A. Kennedy, "Channel, phase noise, and frequency offset in OFDM systems: Joint estimation, data detection, and hybrid Cramer–Rao lower bound," *IEEE Trans. Commun.*, vol. 62, no. 9, pp. 3311–3325, Sep. 2014.
- [27] G. L. Stüber, *Principles of Mobile Communication*. Berlin, Germany: Springer, 2001, vol. 2, pp. 231–343.
- [28] H. Fu and P.-Y. Kam, "Phase-based, time-domain estimation of the frequency and phase of a single sinusoid in AWGN—the role and applications of the additive observation phase noise model," *IEEE Trans. Inform. Theory*, vol. 59, no. 5, pp. 3175–3188, May 2013.
- [29] X. Yi, W. Shieh, and Y. Tang, "Phase estimation for coherent optical OFDM," *IEEE Photon. Technol. Lett.*, vol. 19, no. 12, pp. 919–921, Jun. 2007.
- [30] W. Shieh and C. Athaudage, "Coherent optical orthogonal frequency division multiplexing," *Electron. Lett.*, vol. 42, no. 10, pp. 587–589, 2006.
- [31] H. Abdzadeh-Ziabari, W.-P. Zhu, and M. Swamy, "Joint maximum likelihood timing, frequency offset, and doubly selective channel estimation for OFDM systems," *IEEE Trans. Veh. Technol.*, vol. 67, no. 3, pp. 2787–2791, Mar. 2018.
- [32] P.-S. Wang and D. W. Lin, "On maximum-likelihood blind synchronization over WSSUS channels for OFDM systems," *IEEE Trans. Signal Process.*, vol. 63, no. 19, pp. 5045–5059, Oct. 2015.
- [33] T. Fusco and M. Tanda, "ML-based symbol timing and frequency offset estimation for OFDM systems with noncircular transmissions," *IEEE Trans. Signal Process.*, vol. 54, no. 9, pp. 3527–3541, Sep. 2006.
- [34] X. Ma, H. Zhang, X. Yao, and D. Peng, "Pilot-based phase noise, IQ mismatch, and channel distortion estimation for PDM CO-OFDM system," *IEEE Photon. Technol. Lett.*, vol. 29, no. 22, pp. 1947–1950, Nov. 2017.
- [35] M. Hazewinkel, *Encyclopedia of Mathematics*. Norwell, MA, USA: Kluwer Academic, 1994.

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